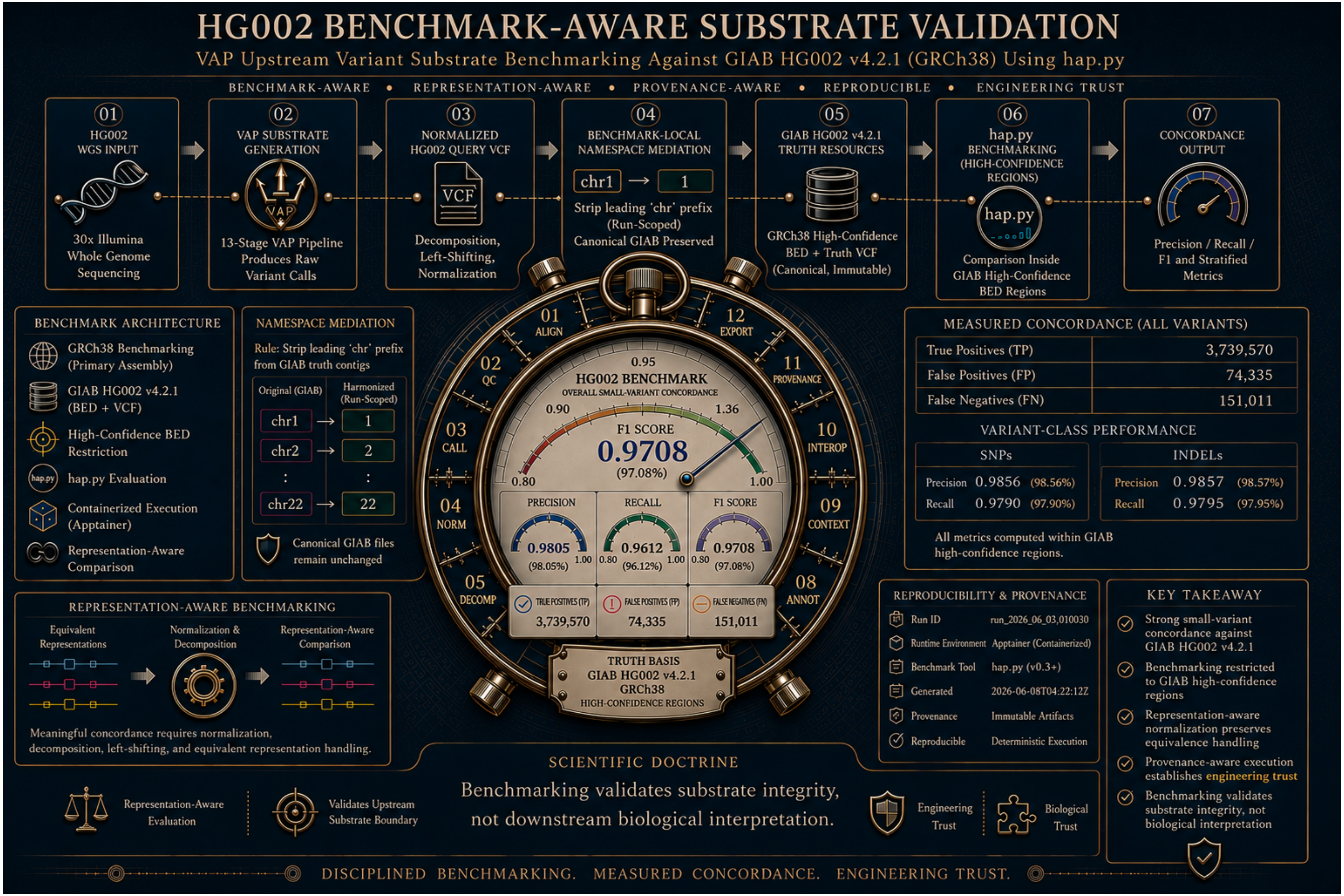
HG002 BENCHMARK-AWARE SUBSTRATE VALIDATION
VAP Upstream Variant Substrate Benchmarking Against GIAB HG002 v4.2.1 (GRCh38) Using hap.py
BENCHMARK-AWARE • REPRESENTATION-AWARE • PROVENANCE-AWARE • REPRODUCIBLE • ENGINEERING TRUST
01 HG002 WGS INPUT
30x Illumina Whole Genome Sequencing
02 VAP SUBSTRATE GENERATION
VAP
13-Stage VAP Pipeline Produces Raw Variant Calls
03 NORMALIZED HG002 QUERY VCF
VCF
Decomposition, Left-Shifting, Normalization
04 BENCHMARK-LOCAL NAMESPACE MEDIATION
chr1 → 1
Strip leading 'chr' prefix (Run-Scoped)
Canonical GIAB Preserved
05 GIAB HG002 v4.2.1 TRUTH RESOURCES
GRCh38 High-Confidence BED + Truth VCF (Canonical, Immutable)
06 hap.py BENCHMARKING (HIGH-CONFIDENCE REGIONS)
hap.py
Comparison Inside GIAB High-Confidence BED Regions
07 CONCORDANCE OUTPUT
Precision / Recall / F1 and Stratified Metrics
BENCHMARK ARCHITECTURE
GRCh38 Benchmarking (Primary Assembly)
GIAB HG002 v4.2.1 (BED + VCF)
High-Confidence BED Restriction
hap.py Evaluation
Containerized Execution (Apptainer)
Representation-Aware Comparison
NAMESPACE MEDIATION
Rule: Strip leading 'chr' prefix from GIAB truth contigs
Original (GIAB)
Harmonized (Run-Scoped)
chr1 → 1
chr2 → 2
: :
chr22 → 22
Canonical GIAB files remain unchanged
01 ALIGN
02 QC
03 CALL
04 NORM
05 DECOMP
08 ANNOT
09 CONTEXT
10 INTEROP
11 PROVENANCE
12 EXPORT
HG002 BENCHMARK
OVERALL SMALL-VARIANT CONCORDANCE
0.95
0.90
1.36
0.80
1.00
F1 SCORE
0.9708
(97.08%)
PRECISION
0.9805
(98.05%)
RECALL
0.9612
(96.12%)
F1 SCORE
0.9708
(97.08%)
TRUE POSITIVES (TP) 3,739,570
FALSE POSITIVES (FP) 74,335
FALSE NEGATIVES (FN) 151,011
TRUTH BASIS
GIAB HG002 v4.2.1
GRCh38
HIGH-CONFIDENCE REGIONS
MEASURED CONCORDANCE (ALL VARIANTS)
True Positives (TP) 3,739,570
False Positives (FP) 74,335
False Negatives (FN) 151,011
VARIANT-CLASS PERFORMANCE
SNPs
Precision 0.9856 (98.56%)
Recall 0.9790 (97.90%)
INDELs
Precision 0.9857 (98.57%)
Recall 0.9795 (97.95%)
All metrics computed within GIAB high-confidence regions.
REPRESENTATION-AWARE BENCHMARKING
Equivalent Representations
Normalization & Decomposition
Representation-Aware Comparison
Meaningful concordance requires normalization, decomposition, left-shifting, and equivalent representation handling.
REPRODUCIBILITY & PROVENANCE
Run ID run_2026_06_03_010030
Runtime Environment Apptainer (Containerized)
Benchmark Tool hap.py (v0.3+)
Generated 2026-06-08T04:22:12Z
Provenance Immutable Artifacts
Reproducible Deterministic Execution
KEY TAKEAWAY
Strong small-variant concordance against GIAB HG002 v4.2.1
Benchmarking restricted to GIAB high-confidence regions
Representation-aware normalization preserves equivalence handling
Provenance-aware execution establishes engineering trust
Benchmarking validates substrate integrity, not biological interpretation
SCIENTIFIC DOCTRINE
Benchmarking validates substrate integrity, not downstream biological interpretation.
Representation-Aware Evaluation
Validates Upstream Substrate Boundary
Engineering Trust
Biological Trust
DISCIPLINED BENCHMARKING. MEASURED CONCORDANCE. ENGINEERING TRUST.